

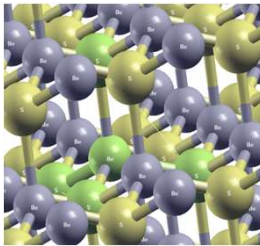
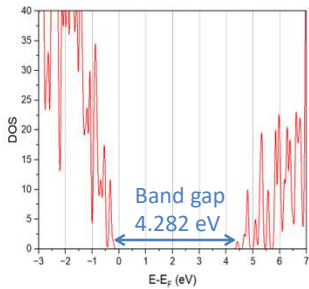
Triplet ground state and optical excitations of local defects in cubic BeS.

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Abstract

Our Density-Functional-Theory, GW, and Bethe-Salpeter calculations show that cubic BeS semiconductor is a promising host for local defects with potential applications in quantum technologies. By applying the postulate that defects leaving under-coordinated at least 3 atoms of relatively high electronegativity and changing the number of electrons by an even number favor triplet ground states, we have found that all 3 studied defects in BeS have a ground triplet state.

Pristine Bulk cBeS



Be: grey balls; S: yellow balls;
Atoms of interest: green balls

cBeS has a zinc-blende structure. As such, it can be determined by one lattice parameter, which we found to be 4.877 Å. (Exp.=4.865 Å). The cohesive energy of cBeS at this lattice parameter is 4.5 eV.

Table 1: Bond lengths around defect (d_{NN}) in the various systems

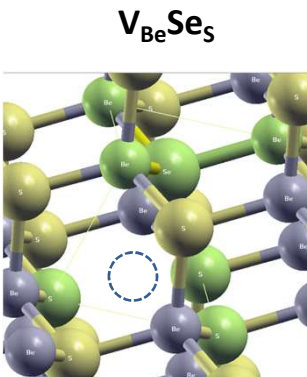
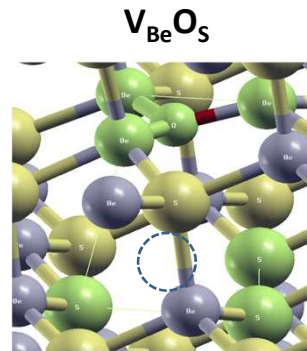
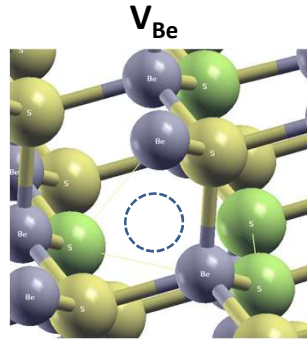
	Bulk	V_{Be}	$V_{Be}O_S$	$V_{Be}Se_S$
$d_{NN}(\text{Å})$	2.1118 (3 S)	2.0753 (3 S)	1.6170 (O) 2.0885 (2 S) 2.0734 (1 S)	2.2164 (Se) 2.0703 (2 S) 2.0818 (1 S)

Conclusions

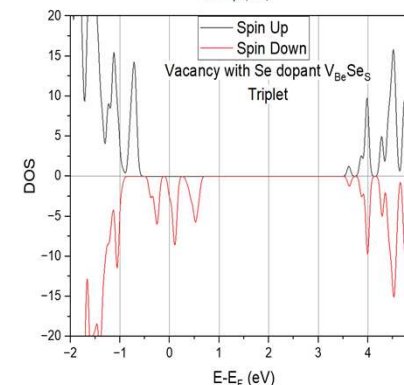
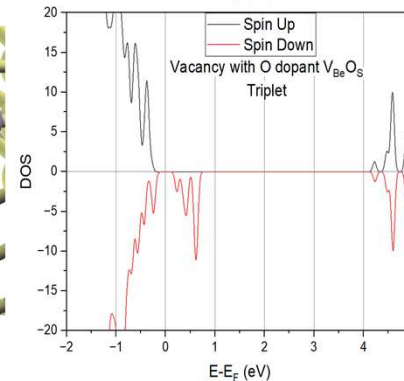
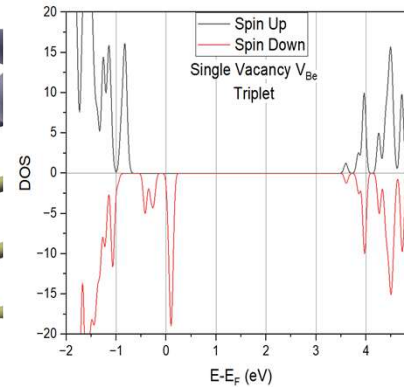
We have found that all 3 studied defects in BeS have a ground triplet state, are dynamically stable, and display intense optical excitations at sufficiently low energies (~ 0.5 eV) such that they may serve as spin qubits that do not require a non-radiative transition to an excited state of the corresponding singlet (still to be found) in order to complete the initialization spin-polarization cycle, but may directly decay to the ground state of the singlet state.

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Crystal Structure



GW DOS



Optical Excitations & Imaginary Part of Dielectric Function

